The Parallel Multiple Mediator Model with Multiple Causal Antecedents

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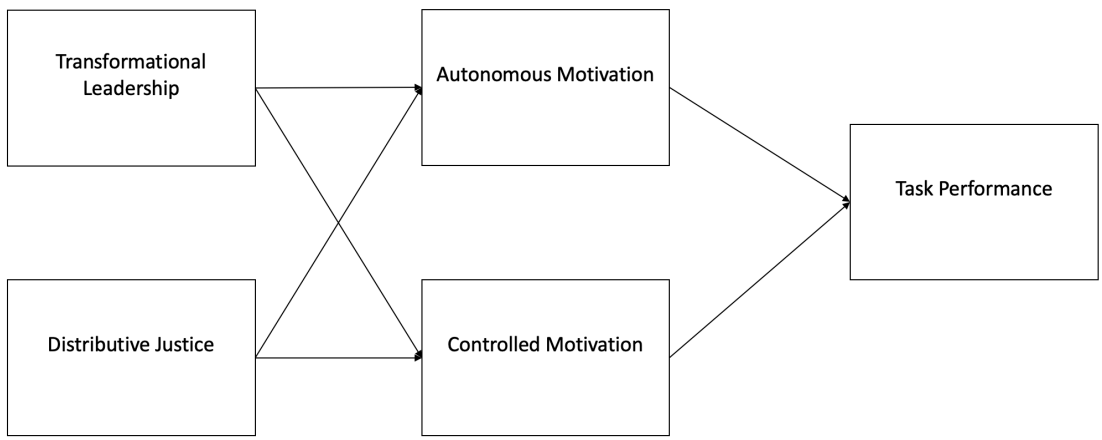
Problem

Why is it that some people have higher task performance whereas others don't?

Theory

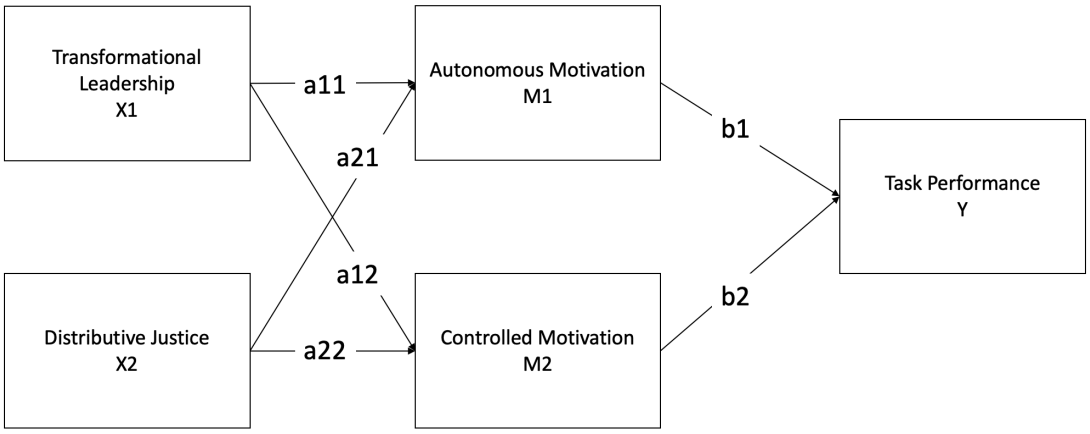
According to self-determination theory, individuals' work behaviors are predicted by their motivations (autonomous and controlled), which, in turn, are predicted by workplace context (e.g., leadership style and distributive justice).



We want to test if this model is true or not. So, we go out to collect some data but what should we do next?

Notation: Diagram

- **X1**: causal antecedent variable1
- **X2**: causal antecedent variable2
- **Y**: consequent variable
- **M1** "mediator 1"
- **M2** "mediator 2"



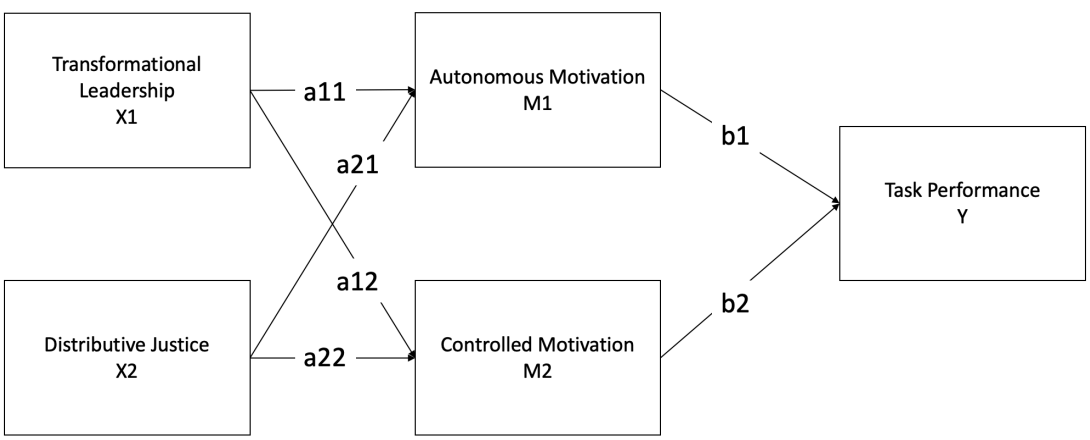
Activity

- Use the notation shown here to draw the relation you would like to investigate.

Notation: Indirect effects

- Let a11, a12, a21, a22, b1, b2, and c' be quantifications of causal effects, such as regression coefficients in an ordinary least squares model (or maximum likelihood path estimates in a structural equation model).
- **a11 × b1 = "indirect effect" of X1 on Y (This is our main focus.)**
- **a21 × b1 = "indirect effect" of X2 on Y (This is our main focus.)**
- **a12 × b2 = "indirect effect" of X1 on Y (This is our main focus.)**
- **a22 × b2 = "indirect effect" of X2 on Y (This is our main focus.)**

Regression Equation



$$AutonomousMotivation_j = i + a11TransformationalLeadership_j + a21DistributiveJustice_j + \varepsilon_j$$

$$ControlledMotivation_j = i + a12TransformationalLeadership_j + a22DistributiveJustice_j + \varepsilon_j$$

$$TaskPerferomance_j = i + b1AutonomousMotivation_j + b2ControlledMotivation_j + \varepsilon_j$$

Transformational Leadership

Transformational leadership is a leadership style that can inspire positive changes in those who follow. Transformational leaders are generally energetic, enthusiastic, and passionate. Not only are these leaders concerned and involved in the process, but they are also focused on helping every member of the group succeed.

Distributive Justice

Distributive justice refers to the ethical principle concerned with the fair allocation of resources, benefits, and burdens among members of a society or group. It focuses on how goods, wealth, opportunities, and responsibilities should be distributed to ensure fairness, equity, and the well-being of individuals.

Run Process Macro

Whenever you would like to run conditional process analysis, don't forget to open and run the script of Process Macro!

```
********************** PROCESS for R Version 4.3.1 **********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

PROCESS is now ready for use.
Copyright 2020-2023 by Andrew F. Hayes ALL RIGHTS RESERVED
Workshop schedule at http://haskayne.ucalgary.ca/CCRAM
```

Load Packages and Import Data

```
library(tidyverse)
library(psych)
library(haven)

data <- read_sav("sdt_2x.sav")

#view our data
data
```

```
# A tibble: 500 × 5
   leader justice    am    cm performance
    <dbl>   <dbl> <dbl> <dbl>       <dbl>
 1   6.16    4.62  5.35  3.14        5.68
 2   3.18    4.09  3.98  4.05        3.73
 3   4.61    4.10  4.43  3.59        4.07
 4   5.02    4.20  4.42  3.92        4.05
 5   4.67    3.93  3.88  3.72        3.88
 6   3.88    3.93  3.80  4.14        3.92
 7   6.38    4.14  4.66  3.74        4.99
 8   3.90    3.74  3.14  3.86        3.41
 9   7       4.22  4.62  3.60        5.10
10   3.95    4.20  4.27  3.37        4.52
# ℹ 490 more rows
```

```r
#descriptive statistics
describe(data)
```

```
            vars   n mean   sd median trimmed  mad  min  max range  skew
leader         1 500 4.00 1.44   3.99    4.00 1.50 1.00 7.00  6.00  0.03
justice        2 500 3.99 0.36   4.00    3.99 0.34 3.01 5.30  2.29  0.12
am             3 500 3.99 0.53   4.00    3.99 0.50 2.32 5.60  3.28 -0.02
cm             4 500 4.01 0.49   4.00    4.00 0.50 2.47 5.54  3.08  0.11
performance    5 500 4.00 0.85   4.01    4.00 0.84 1.14 6.30  5.16 -0.04
            kurtosis   se
leader         -0.53 0.06
justice         0.15 0.02
am              0.14 0.02
cm             -0.09 0.02
performance    -0.04 0.04
```

Visualize (Bivariate Correlations)

```r
data |> as.matrix() |> Hmisc::rcorr()
```

```
            leader justice    am    cm performance
leader        1.00    0.69  0.86 -0.80        0.92
justice       0.69    1.00  0.75 -0.77        0.80
am            0.86    0.75  1.00 -0.77        0.88
cm           -0.80   -0.77 -0.77  1.00       -0.85
performance   0.92    0.80  0.88 -0.85        1.00

n= 500



P
            leader justice am cm performance
leader              0       0  0  0
justice      0              0  0  0
am           0      0          0  0
```

```
cm           0      0       0     0
performance  0      0       0  0

#data |> select(leader, justice, am, cm, performance) |> as.matrix() |> Hmisc
::rcorr()
```

Model (conditional process analysis: the parallel multiple mediator model with multiple causal antecedents)

Unfortunately, there is no existing model we can choose to run the parallel multiple mediator analysis with multiple causal antecedents. However, the conditional process analysis framework allows us to create and customize our own model using bmatrix and cmatrix. This is how:

	X	M1	M2
M1	1		
M2	1	0	
Y	0	1	1
(1, 1, 0,0,1,1)			

	c1
M1	1
M2	1
Y	0
1,1,0	

```
process(data = data, y = "performance", x = "leader", cov = c("justice"), m =
 c("am", "cm"), seed = 12345, bmatrix = c(1, 1, 0, 0, 1, 1), cmatrix = c(1,1,
0))


******************** PROCESS for R Version 4.3.1 ********************
```

```
           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : CUSTOM
    Y : performance
    X : leader
   M1 : am
   M2 : cm

Covariates:
       justice

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: am

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.8878     0.7881     0.0606   924.3644     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     1.2651     0.1444     8.7625     0.0000     0.9815     1.5488
leader       0.2457     0.0106    23.1669     0.0000     0.2249     0.2666
justice      0.4359     0.0425    10.2492     0.0000     0.3524     0.5195

**************************************************************************
Outcome Variable: cm

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.8560     0.7327     0.0634   681.0182     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     7.0069     0.1476    47.4698     0.0000     6.7169     7.2969
leader      -0.1704     0.0108   -15.7144     0.0000    -0.1917    -0.1491
justice     -0.5801     0.0435   -13.3409     0.0000    -0.6656    -0.4947

**************************************************************************
Outcome Variable: performance

Model Summary:
```

```
          R       R-sq        MSE          F        df1        df2          p
     0.9268     0.8589     0.1034 1513.2403     2.0000  497.0000     0.0000

Model:
             coeff         se          t          p       LLCI       ULCI
constant    3.4767     0.3305    10.5184     0.0000     2.8273     4.1261
am          0.8904     0.0419    21.2640     0.0000     0.8082     0.9727
cm         -0.7550     0.0460   -16.4089     0.0000    -0.8454    -0.6646

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******************** ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output: 95

Number of bootstraps for percentile bootstrap confidence intervals: 5000
```

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process(data = data, y = "performance", x = "justice", cov = c("leader"), m =
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```

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**************************************************************************

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    Y : performance
    X : justice
   M1 : am
   M2 : cm

Covariates:
       leader

Sample size: 500

Custom seed: 12345


**************************************************************************
Outcome Variable: am

Model Summary:
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              coeff         se          t          p       LLCI       ULCI
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justice      0.4359     0.0425    10.2492     0.0000     0.3524     0.5195
leader       0.2457     0.0106    23.1669     0.0000     0.2249     0.2666

**************************************************************************
Outcome Variable: cm

Model Summary:
```

```
          R       R-sq        MSE          F        df1        df2          p
     0.8560     0.7327     0.0634   681.0182     2.0000   497.0000     0.0000

Model:
             coeff         se          t          p       LLCI       ULCI
constant    7.0069     0.1476    47.4698     0.0000     6.7169     7.2969
justice    -0.5801     0.0435   -13.3409     0.0000    -0.6656    -0.4947
leader     -0.1704     0.0108   -15.7144     0.0000    -0.1917    -0.1491

**************************************************************************
Outcome Variable: performance

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.9268     0.8589     0.1034  1513.2403     2.0000   497.0000     0.0000

Model:
             coeff         se          t          p       LLCI       ULCI
constant    3.4767     0.3305    10.5184     0.0000     2.8273     4.1261
am          0.8904     0.0419    21.2640     0.0000     0.8082     0.9727
cm         -0.7550     0.0460   -16.4089     0.0000    -0.8454    -0.6646

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```
#Outcome Variable: control => path a12 and path a22

#interpretation: About 73.27% of variance in controlled motivation is explain
ed by (transformational) leadership style and distributive justice.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Leadership significantly and negatively predicts controlled
motivation. One unit increase in leadership is expected to be associated with
 a decrease of 0.51704 units in controlled motivation.

#interpretation: Distributive justice significantly and negatively predicts c
ontrolled motivation. One unit increase in distributive justice is expected t
o be associated with a decrease of 0.5801 units in controlled motivation.

#Outcome Variable: performance => path b1 and path b2

#interpretation: About 85.89% of variance in task performance is explained by
 autonomous motivation AND controlled motivation.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b1: Autonomous motivation significantly and positively predic
ts task performance. One unit increase in autonomous motivation is expected t
o be associated with an increase of 0.8904 units in task performance.

#interpretation b2: Controlled motivation significantly and negatively predic
ts task performance. One unit increase in controlled motivation is expected t
o be associated with an decrease of 0.7550 units in task performance.

#INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a11 x b1 AND a12 x b2
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a11 x b1: Autonomous motivation significantly mediates the re
lation between leadership and task performance (a11b1 = 0.2188, 95% CI [.1923
, .2450]).

#interpretation a11 x b1: Leadership is significantly and indirectly related
to task performance through the pathway of autonomous motivation (a11b1 = 0.2
188, 95% CI [.1923, .2450]).
```

```
#interpretation a12 x b2: Controlled motivation significantly mediates the re
lation between leadership and task performance (a12b2 = 0.1287, 95% CI [.1065
, .1528]).

#interpretation a12 x b2: Leadership is significantly and indirectly related
to task performance through the pathway of controlled motivation (a12b2 = 0.1
287, 95% CI [.1065, .1528]).

#INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a21 x b1 AND a22 x b2
#To evaluate whether M significantly mediates the relation between X and Y, y
ou need to check BootLLCI and BootULCI. If zero is not in the confidence inte
rval, then we can claim an indirect effect different from zero with 95% confi
dence. This is akin to (though not exactly the same as) rejecting the null hy
pothesis of no indirect effect at the alpha = 0.05 level of significance.

#interpretation a21 x b2: Autonomous motivation significantly mediates the re
lation between distributive justice and task performance (a21b2 = 0.3882, 95%
 CI [.3086, .4695]).

#interpretation a21 x b2: Distributive justice is significantly and indirectl
y related to task performance through the pathway of autonomous motivation (a
21b2 = 0.3882, 95% CI [.3086, .4695]).

#interpretation a22 x b2: Controlled motivation significantly mediates the re
lation between distributive justice and task performance (a22b2 = 0.4380, 95%
 CI [.3637, .5158]).

#interpretation a22 x b2: Distributive justice is significantly and indirectl
y related to task performance through the pathway of controlled motivation (a
22b2 = 0.4380, 95% CI [.3637, .5158]).


#Conclusion: When employees perceived their supervisors to be more transforma
tional and rewards are distributed more fairly, the higher autonomous motivat
ion they have, which, in turn, is significantly and positively associated wit
h task performance. Also, when employees perceived their supervisors to be mo
re transformational and rewards are distributed more fairly, the lower contro
lled motivation they have, which, in turn, is significantly and negatively as
sociated with task performance.
```

Assignment 5-1

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) and the income level of child's household (**povlev4_21**) serve as causal antecedent variables, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent

variable. Please run a conditional process analysis to find out the answer. (Let's assume that these six variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 5_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - income level of child's household (**povlev4_21**) : Low values indicate better lower income.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.
  - fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
  - vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
  - teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
- Draw the diagram to represent the paralell multiple mediator model with multiple causal antecedents.
- Write down the bmatrix.
- Write down the cmatrix.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a11, a12, a13, a21, a22, a23, b1, b2, b3, a11 x b1, a12 x b2, a13 x b3, a21 x b1, a22 x b2, and a23 x b3
- Evaluate the overall model and individual predictor.

- Write down your interpretation of the results for each path (i.e., a11, a12, a13, a21, a22, a23, b1, b2, b3, a11 x b1, a12 x b2, a13 x b3, a21 x b1, a22 x b2, and a23 x b3)
- Is your theory correct? Does sugary drink consumption significantly mediate the relation between age and teeth condition and between the household income level and teeth condition. Does fruit consumption significantly mediate the relation between age and teeth condition and between the household income level and teeth condition. Does vegetable consumption significantly mediate the relation between age and teeth condition and between the household income level and teeth condition? Which path is more strongly associated with teeth condition?

Assignment 5-2

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**), the income level of child's household (**povlev4_21**), and their physical activity (**PhysAct_21**) serve as causal antecedent variables, sugary drink consumption (**SugarDrink_21**), fruit consumption (**fruit_21**), and vegetable consumption (**vegetables_21**) serves as parallel mediators, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Please run a conditional process analysis to find out the answer. (Let's assume that these seven variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 5_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - income level of child's household (**povlev4_21**) : Low values indicate better lower income.
  - physical activity (**PhysAct_21**) : High values indicate more physcial activities.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.
  - fruit consumption (**fruit_21**) : Higher values indicate that the child eat more fruits on a weekly basis.
  - vegetable consumption (**vegetables_21**) : Higher values indicate that the child eat more vegetables on a weekly basis.
  - teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
- Draw the diagram to represent the paralell multiple mediator model with multiple causal antecedents.
- Write down the bmatrix.
- Write down the cmatrix.

- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Run Pearson's bivariate correlations
- Run the conditional process analysis
- Find out the effect of path a11, a12, a13, a21, a22, a23, a31, a32, a33, b1, b2, b3, a11 x b1, a12 x b2, a13 x b3, a21 x b1, a22 x b2, a23 x b3, a31 x b1, a32 x b2, and a33 x b3
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a11, a12, a13, a21, a22, a23, a31, a32, a33, b1, b2, b3, a11 x b1, a12 x b2, a13 x b3, a21 x b1, a22 x b2, a23 x b3, a31 x b1, a32 x b2, and a33 x b3)
- Is your theory correct? Does sugary drink consumption significantly mediate the relations between those causal antecedents and teeth condition? Does fruit consumption significantly mediate the relations between those causal antecedents and teeth condition? Does vegetable consumption significantly mediate the relations between those causal antecedents and teeth condition? Which path is more strongly associated with teeth condition?

Assignment: Find an empirical journal article

- Find an empirical article in your field that uses the parallel multiple mediator model with multiple causal antecedents.
- What is the hypothesized model? That is, what is the diagram that visualizes the relations?
- Define Xs Ms, and Y using layperson's terms
- Find out the values for all the path coefficients and indirect effects.
- Find out R^2
- Share your interpretation of the results for each path and indirect effect using layperson's terms
- Are the mediation effects significant? How do you know?
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 5 mins.
- **Submit your PPT slides to the designated folder on iStudy.**